Mini Project 1.2: results

1. Comparison: random vs adversarial vs margined (ms)

1000 measurements per strategy, each call timed with time.perf_counter(). Margined recommendation = overall observed maximum x (1 + margin); default margin +54% (65% utilization budget).

Measured, not proven: no test campaign guarantees the true worst-case path, cache/pipeline state or interference was exercised. The margin is a heuristic and cannot turn a measurement into a sound bound; a static WCET analysis (or hybrid method) is needed for certification.

Function	Control-flow shape	Random max	Adversarial max	Adv / random	Largest sample from	Margined estimat
arly_exit_search	Early-exit branch	0.2291	0.7100	3.10x	adversarial	1.0923
collatz_steps	Data-dependent loop count	0.0379	0.0915	2.41x	adversarial	0.1408
classify_batch	Nested conditionals	0.2463	0.8391	3.41x	adversarial	1.2909
fir_filter	Fixed control flow	0.2888	0.5525	1.91x	adversarial	0.8500

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2. Full statistics per strategy (ms)

Outlier flag: YES when max > 1.5 x the 99th percentile (a rare spike worth investigating, often OS noise rather than the code path).

Function	Strategy	Mean	Std	Min	Max	p90	p95	p99	Max/p99	Outlier
early_exit_search	random	0.0666	0.0407	0.0009	0.2291	0.1194	0.1274	0.1678	1.36	no
early_exit_search	adversarial (key absent)	0.1383	0.0273	0.1226	0.7100	0.1688	0.1831	0.2125	3.34	YES
collatz_steps	random	0.0065	0.0038	0.0010	0.0379	0.0108	0.0123	0.0183	2.07	YES
collatz_steps	adversarial (n = 77031)	0.0230	0.0068	0.0193	0.0915	0.0329	0.0340	0.0501	1.83	YES
classify_batch	random	0.0968	0.0266	0.0677	0.2463	0.1357	0.1495	0.1939	1.27	no
classify_batch	adversarial (all heavy branch)	0.3145	0.0336	0.2850	0.8391	0.3473	0.3731	0.4403	1.91	YES
fir_filter	random	0.0704	0.0135	0.0622	0.2888	0.0771	0.0907	0.1160	2.49	YES
fir_filter	adversarial (extreme values)	0.0724	0.0195	0.0624	0.5525	0.0877	0.1062	0.1195	4.62	YES

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3. Safety margin versus utilization budget

Keeping utilization below a budget U is equivalent to inflating C by 1/U - 1 (the 60-70% convention of Section 1.7). margin_pct in wcet_tool.estimate() overrides this default.

Utilization budget	Equivalent margin on C
60%	+67%
65%	+54%
70%	+43%